OSCAR CARRILLO

(813) 863-3279 ◇ Tampa, FL
oscarc1@usf.edu ◇ oscarc727.github.io ◇ github.com/oscarc727

EDUCATION

University of South Florida, Judy Genshaft Honors College **May 2029**
Bachelors of Science in Electrical Engineering *GPA: 4.00*

SKILLS

Programming	Verilog, C++, Python, MATLAB
Platforms and Tools	Icarus Verilog, Linux, Git, SolidWorks, Simulink
Hardware	Jetson Orin Nano, FPGAs, ESP32, THT/SMD Soldering

TECHNICAL EXPERIENCE

Lower-Limb Exoskeleton Controls Lead **August 2026 - Present**

- Lead for the control system of a wearable robotic leg exoskeleton that assists hip and knee movement, powered by four AK80-9 motors, an NVIDIA Jetson Orin Nano, and Teensy sensor boards on a CAN bus
- Designing a gait estimation system that detects each phase of a person's walk using force sensitive resistors and a state machine, while logging IMU motion data for future improvement to the estimation system
- Running PID workshops for the 8+ member controls team and leading a hands-on inverted pendulum build, giving newer members practice on real hardware before they work on the exoskeleton

PROJECTS

N-bit ALU in Verilog **October 2026**

- Designed a parameterized combinational Arithmetic Logic Unit with 8 operations (add, subtract, AND, OR, XOR, shift left/right, NOT) and carry, signed overflow, zero, and negative flags
- Implemented subtraction using two's complement, with overflow detected from operand and result sign bits
- Wrote automated testbenches in Icarus Verilog that checked over 1.5 million cases, including every possible 8-bit input (524,288 cases), 1,000,000 random cases and 392 directed corner cases with zero failures
- Set up a free, open-source FPGA toolchain (Yosys, nextpnr, openFPGALoader) to program a Tang Nano 20K board and confirmed it working with an LED test design

Self-Stabilizing PID Spoon **September 2026**

- Wrote a PID control loop for a spoon that keeps itself level to help people with hand tremors eat, using an ESP32, an MPU6050 motion sensor, and two servo motors for actuation
- Tested and tuned the PID gains on the prototype to get fast, stable leveling with minimal overshoot
- Moving the design onto a custom PCB designed by a teammate and porting the code to an STM32 microcontroller

Real-Time Audio Spectrum Analyzer **September 2026**

- Co-built a device that takes an analog audio input and shows live sound frequencies as bars on a color screen, using a Teensy 4.0 and FFT signal processing
- Wrote and debugged the code that turns the analog audio into the on-screen display, and tested the full system